Introduction to Trigonometry — Marking Scheme

Maximum Marks: 40 Duration: 1 hr 30 min

Sections A and B: 1 mark for each correct option.

1. (C) 1 **[1]**

   $\sin 30° + \cos 60° = \frac{1}{2} + \frac{1}{2} = 1.$

2. (B) 1 **[1]**

   $\cot 45° = 1.$

3. (A) 1 **[1]**

   $\sec 60° - \tan 45° = 2 - 1 = 1.$

4. (B) 1 **[1]**

   $\sin^2 \theta + \cos^2 \theta = 1.$

5. (B) $\sec^2 A$ **[1]**

   $1 + \tan^2 A = \sec^2 A.$

6. (D) cosec 0° **[1]**

   cosec 0° is not defined because $\sin 0° = 0.$

7. (B) **[1]**

   $\sin A = \frac{BC}{AC} = \frac{5}{13}.$

8. (B) 30° **[1]**

   $2 \sin \theta = 1$, so $\sin \theta = \frac{1}{2}$; hence $\theta = 30°.$

9. (C) 55° **[1]**

   $\cos A = \sin 35° = \cos 55°$; hence $A = 55°.$

10. (B) **[1]**

    $3 \tan^2 30° + 2 \cos^2 60° = 3 \times \frac{1}{3} + 2 \times \frac{1}{4} = \frac{3}{2}.$

11. (A) **[1]**

Both statements are true. Since opposite side < hypotenuse, their positive ratio sin A is less than 1; R explains A.

12. (D) **[1]**

The assertion is false: tan 60° = √3, not 3. The stated ratio relationship is true.

13. (B) **[1]**

Both statements are true. However, the value of sin 30° does not explain the values of tan 45° and cot 45°.

Section C — Step-wise solutions

14. Ratios in triangle PQR **[2 marks: 1 + 1]**

$PR^2 = PQ^2 + QR^2 = 8^2 + 15^2 = 289$; PR = 17 cm.

$\tan P = \frac{QR}{PQ} = \frac{15}{8}$; $\sec P = \frac{PR}{PQ} = \frac{17}{8}$.

Award the remaining 1 mark: one-half for each correct ratio.

15. Standard-angle evaluation **[2 marks: 1 + 1]**

$2 \times \frac{1}{2} \times \frac{1}{2} + 3 \times 1 - 2^2$

$= \frac{1}{2} + 3 - 4 = -\frac{1}{2}$.

16. Complementary angles **[2 marks: 1 + 1]**

cos 2A = sin (90° − 2A); hence A = 90° − 2A.

3A = 90°; therefore A = 30°.

17. Identity-based simplification **[2 marks: 1 + 1]**

$(1 - \cos^2\theta)\,\mathrm{cosec}^2\theta = \sin^2\theta\,\mathrm{cosec}^2\theta = 1$.

$1 + \cot^2\theta - \mathrm{cosec}^2\theta = \mathrm{cosec}^2\theta - \mathrm{cosec}^2\theta = 0$.

Section D — Step-wise solutions

18. Ratios from cos θ **[3 marks: 1 + 1 + 1]**

$\sin^2\theta = 1 - \frac{49}{625} = \frac{576}{625}$; $\sin\theta = \frac{24}{25}$.

$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{24}{7}$.

$\sec\theta(1 - \sin\theta) = \frac{25}{7} \times (1 - \frac{24}{25}) = \frac{1}{7}$.

19. Product identity and evaluation **[3 marks: 1 + 1 + 1]**

$(\sec\theta - 1)(\sec\theta + 1) = \sec^2\theta - 1$.

$\sec^2\theta - 1 = \tan^2\theta$. Hence proved.

At $\theta = 60°$: $\tan^2 60° = (\sqrt{3})^2 = 3$.

20. Difference of two fractions **[3 marks: 1 + 1 + 1]**

$\text{LHS} = \frac{\cos\theta(1 + \sin\theta) - \cos\theta(1 - \sin\theta)}{1 - \sin^2\theta}$.

$= \frac{2\sin\theta\cos\theta}{\cos^2\theta} = \frac{2\sin\theta}{\cos\theta}$.

$= 2\tan\theta = \text{RHS}$. Hence proved.

The denominators are non-zero because θ is acute.

Section E — Step-wise solutions

21. Right triangle ABC **[5 marks: 1 + 1 + 1 + 1 + 1]**

(a) $AC^2 = AB^2 + BC^2 = 9^2 + 12^2 = 225$; AC = 15 cm.

(b) $\sin A = \frac{BC}{AC} = \frac{12}{15} = \frac{4}{5}$.

$\cos A = \frac{AB}{AC} = \frac{9}{15} = \frac{3}{5}$.

(c) $\text{LHS} = (\sin A + \cos A)^2 = (\frac{4}{5} + \frac{3}{5})^2 = \frac{49}{25}$.

$\text{RHS} = 1 + 2 \times \frac{4}{5} \times \frac{3}{5} = 1 + \frac{24}{25} = \frac{49}{25}$.

Thus LHS = RHS; the identity is verified.

22. Combined standard-angle and ratio problem **[5 marks: 1 + 1 + 1 + 1 + 1]**

(a) $\text{Numerator} = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2} = 1$.

Denominator = 1 + 1 = 2; required value = $\frac{1}{2}$.

(b) $\tan \theta = \frac{4}{3}$; take opposite = 4k and adjacent = 3k.

Hypotenuse = 5k; $\sin \theta = \frac{4}{5}$ and $\cos \theta = \frac{3}{5}$.

$\text{Required value} = \frac{3 \times 4 + 2 \times 3}{3 \times 4 - 2 \times 3} = \frac{18}{6} = 3$.

Accept any mathematically correct alternative method with equivalent steps.